LEAF HEALTH CHECK

Team ID : SWTID-2026-8262

Team Size : 5

Team Leader : RAGAVENDRA D (6177@velsrscollege.com)

Team member : GOPINATH B

(6120@velsrscollege.com)

Team member : MOHANAKRISHNAN S

(6162@velsrscollege.com)

Team member : AVINASH D

(6088@velsrscollege.com)

Team member : VISHVA R

(6233@velsrscollege.com)

Pre-requisites

Before running the Leaf Health Check application, several software tools and libraries must be installed to ensure the system operates correctly.

1. Python 3.8 or Higher

Python is the main programming language used to develop the project.
It provides the environment required to run the Streamlit application and perform image processing tasks.

Python can be downloaded from the official website:
https://www.python.org

2. Streamlit Framework

Streamlit is used to build the interactive web interface for the application.
It allows users to upload images, view analysis results, and download reports directly from a browser.

Documentation:
https://docs.streamlit.io

3. Google AI Studio Account (Optional for AI Diagnosis)

The system uses Gemini AI to generate explanations and recommendations about plant diseases.
To enable this feature, users must create an account in Google AI Studio and generate an API key.

https://aistudio.google.com

4. Required Python Libraries

The application uses several Python libraries for image processing and data analysis. These libraries are included in the requirements.txt file.

Main libraries include:

- **Streamlit**
- **OpenCV**
- **NumPy**
- **Pillow**
- **Google Generative AI API**

Users can install all dependencies using the following command:

pip install -r requirements.txt

Project Workflow

The development and execution of the Leaf Health Check project follows a structured workflow that includes AI initialization, core functionality development, application integration, user interface development, and system testing.

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1. AI Setup & Initialization

Activity 1.1: Create AI Provider Account

To enable AI-based diagnosis, the system uses Google Gemini AI.

Steps:

1. Visit the Google AI Studio website
2. Sign in using a Google account
3. Create an API key
4. Store the API key securely for later use in the application.

Activity 1.2: Configure Environment Variables

The API key should not be stored directly in the code.
Instead, it should be stored in an environment file.

Example .env file:

GEMINI_API_KEY=your_api_key_here

The application loads this key securely during execution.

Activity 1.3: Test AI Connectivity

Before integrating the AI module into the application, the connection to the Gemini API should be tested.

The system sends a sample prompt to verify whether the API responds correctly.

Expected result:

- **AI returns a short explanation for a given plant condition.**

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2. Core Functionalities Development

This stage focuses on building the main modules responsible for plant leaf analysis.

Activity 2.1: Project Structure Design

The project follows a modular structure where each component performs a specific task.

Example structure:

```
Leaf-Health-Check
│
├── app.py
├── image_processing.py
├── severity.py
├── recommendation.py
├── ai_diagnosis.py
├── requirements.txt
└── README.md
```

This structure improves readability and maintainability.

Activity 2.2: Image Processing Module

The image processing module prepares uploaded images for analysis.

Key operations include:

- **Image resizing**
- **Grayscale conversion**
- **Noise reduction**
- **Thresholding**

These techniques improve the detection of damaged areas on leaves.

Activity 2.3: Damage Detection

The system analyzes pixel intensity differences to identify damaged regions such as:

- Dark spots
- Brown patches
- Discolored areas

The algorithm calculates the percentage of affected leaf area.

Activity 2.4: Severity Classification

The system categorizes plant health conditions into five levels:

Affected Area	Severity Level
0–10%	Healthy
11–30%	Mild
31–60%	Moderate
61–80%	Severe
81–100%	Dying

This classification provides users with a quick understanding of plant health.

Activity 2.5: Recommendation Module

The recommendation module provides plant care suggestions.

Examples include:

- **Maintain proper watering schedule**
- **Ensure adequate sunlight exposure**
- **Remove infected leaves**
- **Apply appropriate fertilizer**
- **Improve air circulation around plants**

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3. Backend Implementation (Streamlit Application)

The Streamlit application serves as the main entry point of the system.

Activity 3.1: File Upload Interface

The application allows users to upload plant leaf images using the Streamlit file uploader component.

Supported formats include:

- **JPG**
- **PNG**

- JPEG

Activity 3.2: Image Preview

After uploading an image, the system displays a preview to confirm that the correct image has been selected.

Activity 3.3: Data Processing

Once the image is uploaded:

1. The image processing module analyzes the leaf.
2. The severity module calculates the affected area.
3. The AI diagnosis module generates explanations.
4. The recommendation module suggests plant care actions.

Activity 3.4: Error Handling

To ensure stability, the system includes error handling features such as:

- Invalid file format detection
- Empty image handling
- API failure fallback logic

These features prevent the application from crashing.

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4. User Interface Development

The user interface is built using Streamlit and is designed to be simple and intuitive.

Activity 4.1: Layout Design

The interface includes the following sections:

- Application title and description
- Image upload area
- Image preview
- Diagnosis results
- Detailed analysis
- Recommendations
- Download report option

Activity 4.2: Result Display

The application dynamically displays results such as:

- Affected area percentage
- Severity level
- AI diagnosis
- Plant care recommendations

These results are organized in a clear and readable format.

Activity 4.3: User Experience Improvements

Several enhancements improve the user experience:

- Loading indicators while analyzing images
- Progress bars showing affected area
- Colored severity indicators
- Structured report output

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5. Testing & Optimization

This phase ensures that the application works correctly under different conditions.

Activity 5.1: Functional Testing

Various test scenarios are used to validate the system.

Test cases include:

- Healthy leaf images
- Diseased leaf images
- Low-light images
- Images with background noise

Expected result:

- Accurate detection of damaged regions.

Activity 5.2: Quality Evaluation

Performance is evaluated using the following metrics:

Metric	Evaluation
Damage Detection Accuracy	High
Severity Classification	High
Recommendation Quality	Medium–High
User Interface Usability	High
Response Speed	High

Activity 5.3: Performance Optimization

Several improvements were implemented to enhance performance:

- Image resizing before processing
- Efficient pixel analysis algorithms
- Reduced processing time
- Improved UI responsiveness

Project Description and Use Case Scenarios

Leaf Health Check is an intelligent plant health monitoring system designed to assist users in identifying potential plant diseases through image-based analysis. The application is developed using **Python and the Streamlit framework**, combining computer vision techniques with artificial intelligence to analyze plant leaves and provide useful health insights.

Plant diseases are one of the major challenges faced by farmers, gardeners, and agricultural researchers. Early detection of plant diseases is essential to prevent crop damage, improve productivity, and maintain plant health. However, identifying plant diseases manually often requires expert knowledge and experience. The Leaf Health Check system aims to simplify this process by providing an automated solution that can analyze plant leaf images and identify possible signs of damage or infection.

The application allows users to upload an image of a plant leaf through an intuitive web-based interface. Once the image is uploaded, the system performs several image preprocessing steps to prepare the image for analysis. These preprocessing techniques include resizing the image to a standard resolution, converting the image into grayscale, reducing noise using smoothing filters, and applying thresholding techniques to highlight damaged or discolored regions of the leaf.

After preprocessing, the system analyzes the image to identify abnormal patterns such as discoloration, dark spots, and damaged edges. These patterns often indicate plant diseases or nutrient deficiencies. The system calculates the percentage of affected leaf area by comparing the number of damaged pixels with the total number of

pixels in the leaf region. This percentage is used to estimate the severity of the plant condition.

Based on the calculated affected area, the system classifies the plant health condition into different severity categories. These categories include **Healthy, Mild, Moderate, Severe, and Dying**. This classification helps users quickly understand the seriousness of the plant disease and determine whether immediate action is required.

To enhance the diagnostic capability of the system, the application integrates **AI-based analysis using Google's Gemini AI**. The AI module interprets the detected severity level and provides possible explanations for the plant condition. It generates human-readable descriptions of potential plant diseases and suggests appropriate treatment actions.

In addition to disease detection, the system also provides **plant care recommendations**. These recommendations are designed to help users improve plant health and prevent further damage. Suggested actions may include adjusting watering schedules, increasing sunlight exposure, improving airflow around plants, removing infected leaves, and applying suitable fertilizers or treatments.

Use Case Scenarios

The Leaf Health Check application can be used in various real-world scenarios where plant health monitoring is required. The system provides a quick and automated way to analyze plant conditions without requiring expert knowledge.

Scenario 1: Leaf Image Upload and Analysis

A user opens the Leaf Health Check application and uploads an image of a plant leaf using the provided file upload interface. The system accepts image formats such as JPG, PNG, or JPEG. Once the image is uploaded, the system displays a preview of the image and prepares it for analysis. The image is processed using several computer vision techniques to highlight damaged regions and remove unnecessary noise.

Scenario 2: Detection of Damaged Leaf Areas

After preprocessing the image, the system analyzes the leaf to detect damaged or discolored regions. These regions may represent fungal infections, nutrient deficiencies, or other plant health issues. The system identifies darker areas or abnormal color variations and calculates the percentage of the leaf that appears damaged.

Scenario 3: Severity Classification

The system evaluates the percentage of affected leaf area and assigns a severity level to the plant condition. If the damaged area is very small, the plant is classified as Healthy or Mild. If a large portion of the leaf is damaged, the system classifies the condition as Moderate, Severe, or

Dying. This classification helps users quickly understand the urgency of the plant health issue.

Scenario 4: AI-Based Disease Explanation

Once the severity level is determined, the Gemini AI module generates a detailed explanation of possible plant diseases or causes of damage. The AI system provides natural language descriptions that help users understand what might be affecting their plant.

Scenario 5: Plant Care Recommendations

Based on the severity and AI diagnosis, the system generates practical plant care suggestions. These suggestions help users improve plant health and prevent disease spread. Recommendations may include watering adjustments, sunlight management, fertilizer usage, and removal of infected leaves.

Scenario 6: Report Generation and Download

After completing the analysis, the system generates a detailed plant health report. This report includes the percentage of affected area, severity classification, AI diagnosis results, and recommended plant care tips. Users can download this report as a text file for future reference.

Conclusion

Leaf Health Check provides an intelligent solution for plant health monitoring by combining image processing and artificial intelligence technologies. The system enables users to quickly analyze plant leaves and identify potential health issues through a simple and user-friendly interface.

By leveraging computer vision techniques, the application detects damaged regions and calculates the severity of plant diseases. The integration of Gemini AI further enhances the system by generating detailed explanations and treatment suggestions.

The modular architecture of the application ensures scalability and maintainability, allowing future improvements such as deep learning-based plant disease detection and mobile integration.

Leaf Health Check demonstrates how artificial intelligence can assist farmers, gardeners, and plant enthusiasts in maintaining plant health and preventing crop damage. By simplifying disease detection and providing actionable recommendations, the system helps users make informed decisions about plant care and agricultural management.